

River Test

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Ferry travelling across a river(20 points)

1. (4 points) A ferry is crossing a river that is 18m wide at a velocity of 9 ms^{-2} right relative to the water. However, the water is travelling at 3 ms^{-2} down relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
2. (4 points) A ferry is crossing a river that is 168m wide at a velocity of 24 ms^{-2} right relative to the water. However, the water is travelling at 8 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
3. (4 points) A ferry is crossing a river that is 24m wide at a velocity of 24 ms^{-2} right relative to the water. However, the water is travelling at 4 ms^{-2} down relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
4. (4 points) A ferry is crossing a river that is 8m wide at a velocity of 1 ms^{-2} right relative to the water. However, the water is travelling at 7 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
5. (4 points) A ferry is crossing a river that is 24m wide at a velocity of 4 ms^{-2} right relative to the water. However, the water is travelling at 5 ms^{-2} down relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat

Ferry travelling straight across a river(15 points)

6. (3 points) A boat is crossing a river that is 18m wide with a velocity of 2 ms^{-2} relative to the water. However, the water is travelling at 1 ms^{-2} down relative to the ground. If the boat wants to travel in a straight line across, calculate:
- (a) The resultant velocity of the boat relative to the ground
 - (b) The required angle the boat needs to travel at to cross the river directly across
 - (c) The time taken to cross the river
7. (3 points) A boat is crossing a river that is 35m wide with a velocity of 5 ms^{-2} relative to the water. However, the water is travelling at 1 ms^{-2} up relative to the ground. If the boat wants to travel in a straight line across, calculate:
- (a) The resultant velocity of the boat relative to the ground
 - (b) The required angle the boat needs to travel at to cross the river directly across
 - (c) The time taken to cross the river
8. (3 points) A boat is crossing a river that is 72m wide with a velocity of 9 ms^{-2} relative to the water. However, the water is travelling at 1 ms^{-2} down relative to the ground. If the boat wants to travel in a straight line across, calculate:
- (a) The resultant velocity of the boat relative to the ground
 - (b) The required angle the boat needs to travel at to cross the river directly across
 - (c) The time taken to cross the river
9. (3 points) A boat is crossing a river that is 60m wide with a velocity of 15 ms^{-2} relative to the water. However, the water is travelling at 9 ms^{-2} down relative to the ground. If the boat wants to travel in a straight line across, calculate:
- (a) The resultant velocity of the boat relative to the ground
 - (b) The required angle the boat needs to travel at to cross the river directly across
 - (c) The time taken to cross the river
10. (3 points) A boat is crossing a river that is 9m wide with a velocity of 9 ms^{-2} relative to the water. However, the water is travelling at 4 ms^{-2} up relative to the ground. If the boat wants to travel in a straight line across, calculate:
- (a) The resultant velocity of the boat relative to the ground
 - (b) The required angle the boat needs to travel at to cross the river directly across
 - (c) The time taken to cross the river